

gilmore-lie-groups: residuals

equations measured 2026-09-05T11:57:22Z against the model of 2026-09-03

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E09584 (11.4) [conf 0.001]	188	■ 0.001 ● C +67	$\left.\left.\left.\left[\begin{array}{cccccccc} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & \ddots & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & -1 & \\ & & & & & & & 1 \\ & & & & & & & & -1 & \\ & & & & & & & & & \ddots \end{array}\right]\right.\right.\right.$ $\begin{array}{c} H_1 \\ H_2 \\ H_3 \\ \vdots \\ H_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \\ \vdots \end{array}$	(not rendered)	
gilmore-lie-groups_ E09778	243	■ 0.264 ●	$\begin{array}{l} \langle l' L_- l \rangle \langle \theta \phi L_- \theta' \phi' \rangle \\ \langle \theta \phi L_- \theta' \phi' \rangle \\ \langle \theta \phi L_- \theta' \phi' \rangle \end{array}$ Matrix element of the angular momentum shift down operator in the coordinate representation: $\hbar(-\partial/\partial\theta + i(\cos\theta/\sin\theta)(\partial/\partial\phi))\delta(\cos\theta' - \cos\theta)\delta(\phi' - \phi)$. Matrix element of the angular momentum shift down operator in the algebraic representation: $\hbar\sqrt{(l' - m')(l' + m)}$. Matrix element of the similarity transformation between the coordinate representation and algebraic representation. Also called spherical harmonic: $Y_m^l(\theta, \phi)$.	$\langle \theta \phi L_- \theta' \phi' \rangle$ $\langle l' L_- l \rangle$ $\langle \theta \phi L_- \theta' \phi' \rangle$	Matrix element of the angular momentum shift down operator in the coordinate representation: $\hbar(-\partial/\partial\theta + i(\cos\theta/\sin\theta)(\partial/\partial\phi))\delta(\cos\theta' - \cos\theta)\delta(\phi' - \phi)$. Matrix element of the angular momentum shift down operator in the algebraic representation: $\hbar\sqrt{(l' - m')(l' + m)}$. $\delta_{l'l}\delta_{m',m-1}$. Matrix element of the similarity transformation between the coordinate representation and algebraic representation. Also called spherical harmonic: $Y_m^l(\theta, \phi)$.
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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gilmore-lie-groups_ E00381 (7.19)	119	■ 0.448 ●	$\begin{array}{ccc} \frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] & + & \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right] \\ & \downarrow & \downarrow \text{EXP} \\ \text{EXP} \downarrow & & \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right] \end{array}$	$\frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right]$ $\text{EXP} \downarrow \quad \downarrow \quad \downarrow \text{EXP}$ $[SU(1, 1)/U(1)] \quad \times \quad \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right]$	
gilmore-lie-groups_ E00590 (11.11)	190	■ 0.454 ● C -38	$\begin{array}{ccc} \frac{\mathfrak{su}(2)}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] & = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} \mathfrak{su}(2) - \mathfrak{u}(1) & \\ a_1 + ia_2 & a_1 - ia_2 \end{array} \right] & \mathfrak{u}(1) \end{array}$	$\frac{\mathfrak{su}(2)}{2} = \mathfrak{u}(1) + \mathfrak{su}(2) - \mathfrak{u}(1)$ $\frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$	
gilmore-lie-groups_ E00369 (7.7)	115	■ 0.468 ●	$\begin{array}{l} \operatorname{EXP} \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right] \\ = \left[\begin{array}{cc} \cosh r + a \sinh r/r & (b+c) \sinh r/r \\ (b-c) \sinh r/r & \cosh r - a \sinh r/r \end{array} \right] \quad r^2 = a^2 + b^2 - c^2 > 0 \\ = \left[\begin{array}{cc} 1+a & \\ b-c & \\ 1-a & \end{array} \right] \quad a^2 + b^2 - c^2 = 0 \\ = \left[\begin{array}{cc} \cos r + a \sin r/r & (b+c) \sin r/r \\ (b-c) \sin r/r & \cos r - a \sin r/r \end{array} \right] \quad -r^2 = a^2 + b^2 - c^2 < 0 \end{array}$	$\operatorname{EXP} \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right]$ $= \left[\begin{array}{cc} \cosh r + a \sinh r/r & (b+c) \sinh r/r \\ (b-c) \sinh r/r & \cosh r - a \sinh r/r \end{array} \right] \quad r^2 = a^2 + b^2 - c^2 > 0$ $= \left[\begin{array}{cc} 1+a & b+c \\ b-c & 1-a \end{array} \right] \quad a^2 + b^2 - c^2 = 0$ $= \left[\begin{array}{cc} \cos r + a \sin r/r & (b+c) \sin r/r \\ (b-c) \sin r/r & \cos r - a \sin r/r \end{array} \right] \quad -r^2 = a^2 + b^2 - c^2 < 0$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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gilmore-lie-groups_ E00817 (14.53)	256	0.675	$\begin{array}{rlr} M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & M_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} & M_2^t = -M_2 \\ M_1^t = +M_1 & G_1^t = -G_1 & \\ G_2^t = +G_2 & & \end{array}$	$M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$ $M_1^t = +M_1 \quad G_1^t = -G_1 \quad M_2^t = -M_2 \quad G_2^t = +G_2$	
gilmore-lie-groups_ E00741	232	0.752	$\begin{array}{l} \lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2j\epsilon^2 \rightarrow 1 \\ j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2\epsilon^2} \rightarrow h_3 \quad \left \begin{matrix} j \\ m \end{matrix} \right \rightarrow \\ j + m = n(\text{finite}) \quad \theta \rightarrow \frac{\pi}{2} - \sqrt{2}\epsilon x \end{array}$	$\lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2j\epsilon^2 \rightarrow 1$ $j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2\epsilon^2} \rightarrow h_3 \quad \left \begin{matrix} j \\ m \end{matrix} \right\rangle \rightarrow \left \begin{matrix} \infty \\ n \end{matrix} \right\rangle$ $j + m = n(\text{finite}) \quad \theta \rightarrow \frac{\pi}{2} - \sqrt{2}\epsilon x$	
gilmore-lie-groups_ E00601 (11.22)	193	0.901	$\begin{array}{l} \begin{array}{r} \mathfrak{su}(2) = \frac{1}{2} \left[\begin{array}{cc} ia_3 & ia_1 + a_2 \\ ia_1 - a_2 & -ia_3 \end{array} \right] \rightarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \\ \downarrow \\ \mathfrak{sl}(2; \mathbb{R}) = \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 + a_2 \\ a_1 - a_2 & -a_3 \end{array} \right] \leftarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \\ \downarrow \\ \mathfrak{p} \rightarrow i\mathfrak{p} \downarrow \end{array} \\ \begin{array}{r} \mathfrak{su}(2) = \frac{1}{2} \left[\begin{array}{cc} ia_3 & ia_1 + a_2 \\ ia_1 - a_2 & -ia_3 \end{array} \right] \rightarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \\ \downarrow \\ \mathfrak{sl}(2; \mathbb{R}) = \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 + a_2 \\ a_1 - a_2 & -a_3 \end{array} \right] \leftarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \\ \downarrow \\ \mathfrak{p} \rightarrow i\mathfrak{p} \downarrow \end{array} \end{array}$	$\mathfrak{su}(2) = \frac{1}{2} \left[\begin{array}{cc} ia_3 & ia_1 + a_2 \\ ia_1 - a_2 & -ia_3 \end{array} \right] \rightarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right]$ $\downarrow \quad \mathfrak{p} \rightarrow i\mathfrak{p} \downarrow$ $\mathfrak{sl}(2; \mathbb{R}) = \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 + a_2 \\ a_1 - a_2 & -a_3 \end{array} \right] \leftarrow \frac{1}{2} \left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right]$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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gilmore-lie-groups_ EQ0006 (1.3)	18	0.944	$\begin{array}{r} I \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (123) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] \quad (321) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \\ (12) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (23) \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right] \quad (13) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right] \end{array}$	$I \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (123) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] \quad (321) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right]$ $(12) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (23) \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right] \quad (13) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$	
gilmore-lie-groups_ EQ0113 (3.3)	49	0.991 C -5	$\begin{array}{c} \text{\texttt{\begin{array}{cc} c=(a, b)=a+i b \& a \in \mathbb{R}, b \in \mathbb{R}, i^2=-1 \\ \begin{array}{cc} a & b \\ -b & a \end{array} \longrightarrow \begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array} \end{array}}$	$c = (a, b) = a + ib \quad a \in \mathbb{R}, b \in \mathbb{R}, i^2 = -1$ $(a, b) \longrightarrow \left[\begin{array}{cc} a & b \\ -b & a \end{array} \right] \quad i = (0, 1) \longrightarrow \left[\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array} \right]$	
Conf. MathPix confidence: 0.944 0.991 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C W S N K not measured					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ EQ0624 (1.20)	24	■ 0.222 ● C +21	$\left.\left(r_1-r_2\right)\right\}\rightarrow\left(r_1-r_2\right),$	$(r_1-r_2)\overset{I}{\rightarrow}+(r_1-r_2),$	$(r_1-r_2)\left\{\begin{array}{l}\overset{I}{\rightarrow}+(r_1-r_2)\\ \overset{(12)}{\rightarrow}(r_2-r_1)=-(r_1-r_2)\end{array}\right.$
gilmore-lie-groups_ EQ0474 (8.6)	145	■ 0.233 ● C +114	$\begin{gathered} \eta\left(\hat{n}+\frac{1}{2}\right)+R a^{\dagger 2}+L a^2+r a^{\dagger}+l a+\delta I \\ \mathfrak{d e f}=\left[\begin{array}{cccc} 0 & l & r & -2 \delta \\ 0 & \eta & 2 R & -r \\ 0 & -2 L & -\eta & l \\ 0 & 0 & 0 & 0 \end{array}\right] \\ \mathfrak{R e g}=\left[\begin{array}{ccc ccc} 0 & -2 R & 2 L & -r & l & 0 \\ 4 L & 2 \eta & 0 & 2 l & 0 & 0 \\ -4 R & 0 & -2 \eta & 0 & -2 r & 0 \\ \hline & & & \eta & 2 L & l \\ & & & -2 R & -\eta & -r \\ & & & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I \end{array} \end{gathered}$	(not rendered)	
gilmore-lie-groups_ EQ0556 (10.1)	174	■ 0.447 ● C +25	$\begin{aligned} \left\{\left[H_i, H_j\right]\right\} &=0 \\ \left\{\left[\mathbf{H}, E_{\alpha}\right]\right\} &=\alpha E_{\alpha} \\ \left\{\left[E_{\alpha}, E_{\beta}\right]\right\} &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \end{aligned}$	$\begin{aligned} \left[H_i, H_j\right] &=0 \\ \left[\mathbf{H}, E_{\alpha}\right] &=\alpha E_{\alpha} \\ \left[E_{\alpha}, E_{\beta}\right] &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \end{aligned}$	$\begin{aligned} \left[H_i, H_j\right] &=0 \\ \left[\mathbf{H}, E_{\alpha}\right] &=\alpha E_{\alpha} \\ \left[E_{\alpha}, E_{\beta}\right] &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \quad \alpha+\beta \neq 0 \text { a root} \\ &=0 \quad \alpha+\beta \text { not a root} \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Low confidence

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gilmore-lie-groups_ EQ0249 [conf 0.031]	84	■ 0.031 ●	$\begin{aligned} f_0(\theta) &= \cos \theta & \text{or } f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & \theta^2 f_2(\theta) &= 1 - \cos(\theta) \end{aligned}$	$f_0(\theta) = \cos \theta \quad \text{or } f_0(\theta) = \cos \theta \quad \theta f_1(\theta) = \sin(\theta) \quad \theta^2 f_2(\theta) = 1 - \cos(\theta),$	$\begin{aligned} f_0(\theta) &= \cos \theta & \text{or } f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & \theta^2 f_2(\theta) &= 1 - \cos(\theta) \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					